

A Certified Census of One-Place Stark Invariants over Real Quadratic Fields

Taxonomy, Exact Quadratic Stratum, and the Higher-Order Frontier

Hainan Zhao
Independent Researcher

31 July 2026

Abstract

We give a certified finite census of one-place differenced Stark packets over real quadratic fields for squarefree radicands $2 \leq D \leq 200$ and finite moduli of norm at most 100, with conjugate real places identified. The 8,200 representatives split exactly into 3,936 empty-support rows, 1,560 quadratic-support rows, and 2,704 higher-order rows. The quadratic stratum is solved exhaustively: every row has an exact packet polynomial over its real quadratic base. The higher-order stratum is classified by exact mechanism eligibility and an explicit frontier. These are finite census statements only; we claim no asymptotic density and no new higher-order packet identity.

1 Claim boundary and conventions

Let K be real quadratic and $\mathfrak{m} = f\infty_2$ a one-place modulus. With R the real-place sign class, put

$$Z_{\mathfrak{m}}(s, A) = \zeta_{\mathfrak{m}}(s, A) - \zeta_{\mathfrak{m}}(s, RA), \quad X_A = \exp Z'_{\mathfrak{m}}(0, A).$$

The packet is the Artin-indexed tuple $(X_A)_{A \in \text{Cl}_{\mathfrak{m}}(K)}$. The exact Fourier, Artin-action, place-ordering, and positive-orientation conventions are those of the companion results paper [1]; this paper does not introduce a second convention system.

The proved assertions here are the finite trichotomy, the exhaustive quadratic packet corpus, and exact higher-order mechanism eligibility. We do not assert results beyond the frozen range. Eligibility is not a packet identity. In particular, an Engine-B normal-closure group or a banked representative does not promote another member: exact conductor, ray-class, orientation, and Artin-labelled packet transport are separate obligations. The v5 scope has 232 eligible rows. Its successor ledger records twelve completed noncanonical member transports; the other 220 member transports remain open.

We use *PROVED* for exact replay under checked hypotheses and *OBSERVED* for descriptive statistics of this finite population. “Unconditional” means that an asserted identity follows from a proved theorem after exact hypothesis checks; it does not prove the general Stark conjecture.

2 Frozen universe and trichotomy

The universe consists of maximal-order fields $\mathbb{Q}(\sqrt{D})$ for squarefree $2 \leq D \leq 200$, integral ideals $\mathfrak{f}$ with $N\mathfrak{f} \leq 100$, and the modulus $f\infty_2$. Conjugation exchanges the two real places, so one representative is retained. A clean PARI replay enumerates 121 fields, 13,939 raw ideals, and 8,200 representatives.

Let T be empty differenced Fourier support, Q be nonempty quadratic support, and H be the remaining higher-order support. The structural sign-class lemmas in [1] show that empty support is exactly the trivial packet condition.

Theorem 1 (Frozen trichotomy). *In the frozen universe,*

$$|T| = 3936, \quad |Q| = 1560, \quad |H| = 2704.$$

Every row belongs to exactly one stratum. Every packet in T is one.

The earlier v5 routing declaration recorded 3,899 trivial rows. The remaining 37 are empty-support rows that its obsolete routing order assigned an exponent-cap label before applying the empty-support theorem. The historical record is preserved; Theorem 1 uses the support-first replay.

3 The exact quadratic stratum

For a supported quadratic character, the uniform theorem in [1], based on Tate’s rank-one algebraicity theorem [2, Thm. IV.5.4], gives exact exponents from imprimitive Euler factors and a relative regulator index. Conductors, Euler factors, relative-unit kernels, orientations, and indices are checked exactly. There are 2,232 supported character occurrences: 672 have zero Euler factor, and 346 rows have all factors zero and packet polynomial $X - 1$.

If u_i is a norm-one relative unit, then

$$t_i = u_i + u_i^{-1} = \text{Tr}_{L_i/K}(u_i) \in K.$$

The formal sign-orbit polynomial is therefore synthesized without its compositum by

$$P_0(x) = x - 1, \quad P_i(x) = \text{Res}_z(P_{i-1}(z), x^2 - t_i x z + z^2). \quad (1)$$

Each root z is replaced by zu_i and zu_i^{-1} , so every intermediate polynomial remains over K . The recurrence concerns denominator-cleared packet powers. At denominator two, the positive packet factor is selected only after exact square-lift, reciprocity, irreducibility, positivity, and Artin-image checks.

Theorem 2 (Exhaustive quadratic stratum). *Every one of the 1,560 rows of Q has a certified exact packet polynomial over its base field. The stable RQ-ordered row certificates have terminal hash-chain root*

`7c04242b1d4c11293af96f83f4915dbed25f6125c60d82965e533df5c9d81855.`

Table 1: Exact quadratic-stratum distributions.

invariant	count
packet degree 1, 2, 4, 8 over K	346, 930, 242, 42
common denominator 1, 2	1491, 69
all-Euler-zero packet $X - 1$	346
maximum coordinate height	62 digits

The preregistered height-only predictor had maximum 89 digits, below the frozen 256-digit cap. The full exact run took 43.46 seconds in the pinned environment; this is an OBSERVED engineering statistic. A separate frozen 50-row analytic audit passed 43 nonzero character comparisons and 101 Artin sign rows at 384 bits. That is an independent certified numerical consistency check, not a proof input.

3.1 A worked imprimitive row

RQ-000013 over $\mathbb{Q}(\sqrt{2})$ has finite ideal HNF $[14, 6; 0, 2]$ of norm 28 and ray group C_2 . Its supported character has primitive conductor obtained by removing one prime of norm two. The exact imprimitive Euler factor and relative index are both two:

$$E_\chi = 2, \quad I_\chi = 2.$$

The oriented relative unit has minimal polynomial

$$u^4 - 2u^3 + u^2 - 2u + 1$$

and its selected real root lies in $(9/5, 19/10)$. Exact Artin labels then give

$$X_{[0]} = u^2, \quad X_{[1]} = u^{-2}.$$

This is a PROVED application of the uniform theorem, including the otherwise easy-to-miss nonzero imprimitive Euler branch. A numerical point-value comparison is retained only as an OBSERVED cross-check.

4 Higher-order taxonomy and frontier

For every H row we record Shintani index-two eligibility, cyclic-quartic CM eligibility, exact weak-hypothesis coverage of Roblot’s Theorem 7.1 [7], and any selected certificate in the companion results paper. This is a taxonomy of mechanisms, not a claim that an eligible row has acquired a new packet identity.

Table 2: Exact finite H taxonomy.

category	rows
higher-order rows	2704
Shintani index-two eligible	232
cyclic-quartic CM eligible	881
exclusive frontier	1591
Roblot full weak coverage	1079
Roblot hypothesis failure	405
unsupported support order	1215
incomplete legacy quartic construction	5
all registered mechanisms fail	1359

For order-six support, 764 character/inverse-character kernels reduce to 407 primitive-field keys. Exact local checks remove 25 keys and 382 require field certificates. The residual 73 possible three-divisible class-number fields are each closed by an exact unramified cyclic-cubic certificate. Thus every sextic kernel has an exact Roblot applicability decision. The five incomplete entries of Table 2 are old quartic construction failures, not missing sextic decisions.

The index column uses the genuine derived-subgroup order in the common stable modulus, rather than the legacy W1 screening proxy. This distinction resolves a potentially misleading control case: RQ-005298 has legacy displayed index two, but its certified normal closure has relative degree 128, maximal absolutely abelian subfield of relative degree 32, and therefore derived-subgroup order $128/32 = 4$. It fails the actual Engine-B index-two predicate; its non-eligibility is not an unstated failure of a closure, conductor, or transfer condition. The separately incomplete quartic Engine-C resolvent for this row has no mathematical verdict.

The headline wall is RQ-000692 over $\mathbb{Q}(\sqrt{21})$, with finite norm 36 and support profile (2, 6). Its Shintani index is six. The one-place, local- S , and prime-to-three-class-number conditions of Roblot’s theorem pass, but a prime above 3 is wildly ramified in the relevant sextic extension. The theorem therefore does not apply. This is an exact limitation of the registered methods, not a non-algebraicity theorem.

4.1 Minimal frontier targets

The frontier artifact selects the least RQ identifier at each support order, independently of the outcome of later mechanism checks. A representative initial segment is given in Table 3. The complete table, including orders through 210, is machine-readable in the H-taxonomy artifact. The distinction between “unresolved” and “all mechanisms fail” is retained: an unresolved row can still be eligible for one of the registered mechanisms.

Table 3: First frozen frontier targets by support order.

order	minimal unresolved	minimal all-mechanisms-fail
4	RQ-000030	RQ-000277
6	RQ-000032	RQ-000227
8	RQ-000018	RQ-000018
10	RQ-000078	RQ-000078
12	RQ-000294	RQ-000294
14	RQ-000524	RQ-000524
16	RQ-001056	RQ-001056
18	RQ-000657	RQ-000657
20	RQ-000785	RQ-000785
24	RQ-000363	RQ-000363

5 Engine-B transport scope

The v5 Engine-B population consists of 232 rows in 88 groups by exact normal-closure polynomial. The historical scope had 195 eligible rows and 159 pending member identities. All survive in v5; 37 new eligible rows are added. Its immutable scope manifest records each member, its canonical closure representative, and its frozen finite ideal. A successor ledger promotes twelve noncanonical members. RQ-000039 is transported from RQ-000021 by

$$X_{98}(A) = X_{49}(A)X_{49}(A\mathfrak{q}^{-1})^{-1},$$

where $\mathfrak{q}$ is the unique norm-two prime of $\mathbb{Q}(\sqrt{2})$. The exact ray-class map is the identity on C_6 , including the sign class. The four targets RQ-000195, RQ-000200, RQ-000205, and RQ-000213 are transported from the sealed RQ-000190 packet by exact one- or two-prime Euler-deletion products. This is a distribution relation, not equality of packets. The source-coprime RQ-000221 and RQ-000228 use the exact label conversion $a \mapsto 5a$ in the same source C_6 ; their quotient prime powers still each delete one local Euler factor. Five further B5-022 members are transported from the sealed RQ-000419 packet; three use the exact generator relabelling matrix [5] in the source C_6 . The B5-086 direct-source closure shares source prime 11 in every target enlargement, while B5-021 and B5-033 admit no integral source-to-target modulus quotient. These are exact exclusions from the coprime Euler-deletion route, not negative packet claims. The label-aware extension is tracked separately from the earlier identity-label B5-025 batch. The remaining 220 members retain status `UNSTARTED_NO_CASE_LEVEL_PACKET_CLAIM`.

Six historical rows have a different printed normal-closure polynomial in the v5 reconstruction. Both records are retained. Literal polynomial equality is not used to infer field identity or packet transport. This is why the present paper reports mechanism eligibility separately from case-level higher-order identities.

6 Reproducibility and relation to earlier work

Every material assertion has a versioned JSON artifact, input hashes, and replay code under PARI/GP 2.15.4. The primary finite evidence is the support-first reconciliation, the row-level quadratic corpus, and the H-taxonomy matrix. The companion paper proves the uniform quadratic theorem and selected higher-order identities; this census does not reprove or enlarge those higher-order identities.

Tate supplies the rank-one algebraicity input [2]; Arakawa [3], Dummit–Sands–Tangedal [4, 5], and Cohen–Roblot [6] give computational and index-theoretic antecedents. Roblot’s weak theorem is not claimed as new: Table 2 quantifies its exact coverage in this finite range. The contribution here is a provenance-preserving finite classification and an exhaustive exact quadratic packet corpus.

The prior-art comparison is deliberately narrow. Cohen–Roblot study Hilbert class fields, whereas the present promoted examples and census rows are one-place ray-field objects with nontrivial finite modulus. An exact four-control subfield audit is mixed: the selected ray normal closures for bases 35, 42, and 186 contain the associated Hilbert biquadratic field, while the selected base-51 closure does not. This is a PROVED statement about the four frozen normal closures only; it identifies neither a Stark unit nor an Artin-labelled packet and does not generalize containment. Likewise, Dummit–Sands–Tangedal provide theorem-class antecedents for quadratic-character phenomena; this paper makes no priority assertion from that overlap. The detailed comparison table remains an appendix artifact rather than an input to any census count.

A Record map and replay boundary

The trichotomy is in `artifacts/census-paper-layer0-reconciliation-v1.json`; the quadratic audit is in `artifacts/census-q-packet-corpus-audit-v1.json`; the H matrix is in `artifacts/census-h-taxonomy-v2.json`; the preserved v1 record used the legacy displayed-index proxy and is corrected by the v2 provenance fields. Engine-B transport scope is in `artifacts/engine-b-transport-manifest-v5.json`; its successor ledger is in `artifacts/engine-b-transport-ledger-v4.json`. The latter records exactly twelve completed member transports. The exact direct-source coprime no-go screens are recorded in `artifacts/b5086-transport-geometry-v1.json` and `artifacts/final-direct-source-coprime-screen-v1.json`. The exact four-control Hilbert/ray object audit is in `artifacts/b5079-hilbert-ray-containment-v1.json` and `artifacts/hilbert-ray-containment-tranche-v1.json`; it is field containment only. The project checksum manifest and full test suite replay source hashes and schema gates.

References

- [1] H. Zhao, *Effective Archimedean Stark Theorems over Real Quadratic Fields: Quadratic Support, Shintani Transfer, and CM Descent*, version 1.5, 2026, <https://doi.org/10.5281/zenodo.21713178>.
- [2] J. Tate, *Les conjectures de Stark sur les fonctions L d’Artin en $s = 0$* , Progress in Mathematics 47, Birkhäuser, 1984.

- [3] T. Arakawa, *On the Stark–Shintani conjecture and certain relative class numbers*, Adv. Stud. Pure Math. 7 (1985), 1–6.
- [4] D. S. Dummit, J. W. Sands, and B. A. Tangedal, *Computing Stark units for totally real cubic fields*, Math. Comp. 66 (1997), 1239–1267.
- [5] D. S. Dummit, J. W. Sands, and B. A. Tangedal, *Stark’s conjecture in multiquadratic extensions, revisited*, J. Théorie des Nombres de Bordeaux 15 (2003), 83–97.
- [6] H. Cohen and X.-F. Roblot, *Computing the Hilbert class field of real quadratic fields*, Math. Comp. 69 (2000), 1229–1244.
- [7] X.-F. Roblot, *Index formulae for Stark units and their solutions*, Pacific J. Math. 266 (2013), 391–422.